



ALTER ERA
ENGINEERING



Tidal Shuffle

Engineering and operation manual

A smarter shuffle for the TIDAL app on the Mac. It follows the song you are hearing, chooses a related one, and makes TIDAL play it when the song ends.

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ABOUT IT

It describes Tidal Shuffle 0.12.0: how to install and run it, what each part does, and how to put it right when something goes wrong. A one-page operation card (AE-TS-00) sums up daily use.

CONVENTIONS

- `n` a key to press
- `tidal-shuffle run` something to type in Terminal
- `player.handoff_mode` a setting in the config file
- **Esc** → **Theme** a place in the settings menu
- In the figures, **orange** marks the main path; **green**, a call to a web service.

WHAT IT DOES NOT DO

It never changes your TIDAL library or playlists, never plays through Spotify's speakers (Spotify is muted while it is used), and never interrupts Spotify while you are listening to it.

01 · OVERVIEW

What it is

TIDAL's own shuffle plays your list in a random order. Tidal Shuffle instead listens to what is playing and finds the next song that belongs with it, from Spotify's song radio and other sources, then plays it in TIDAL.

- **Never cuts a song short**

TIDAL is held for the last moment of a song, then the pick starts.

- **Never makes you wait**

Songs are chosen in the background, one ahead.

- **Stays out of the way**

Spotify is hidden and muted while it is asked.

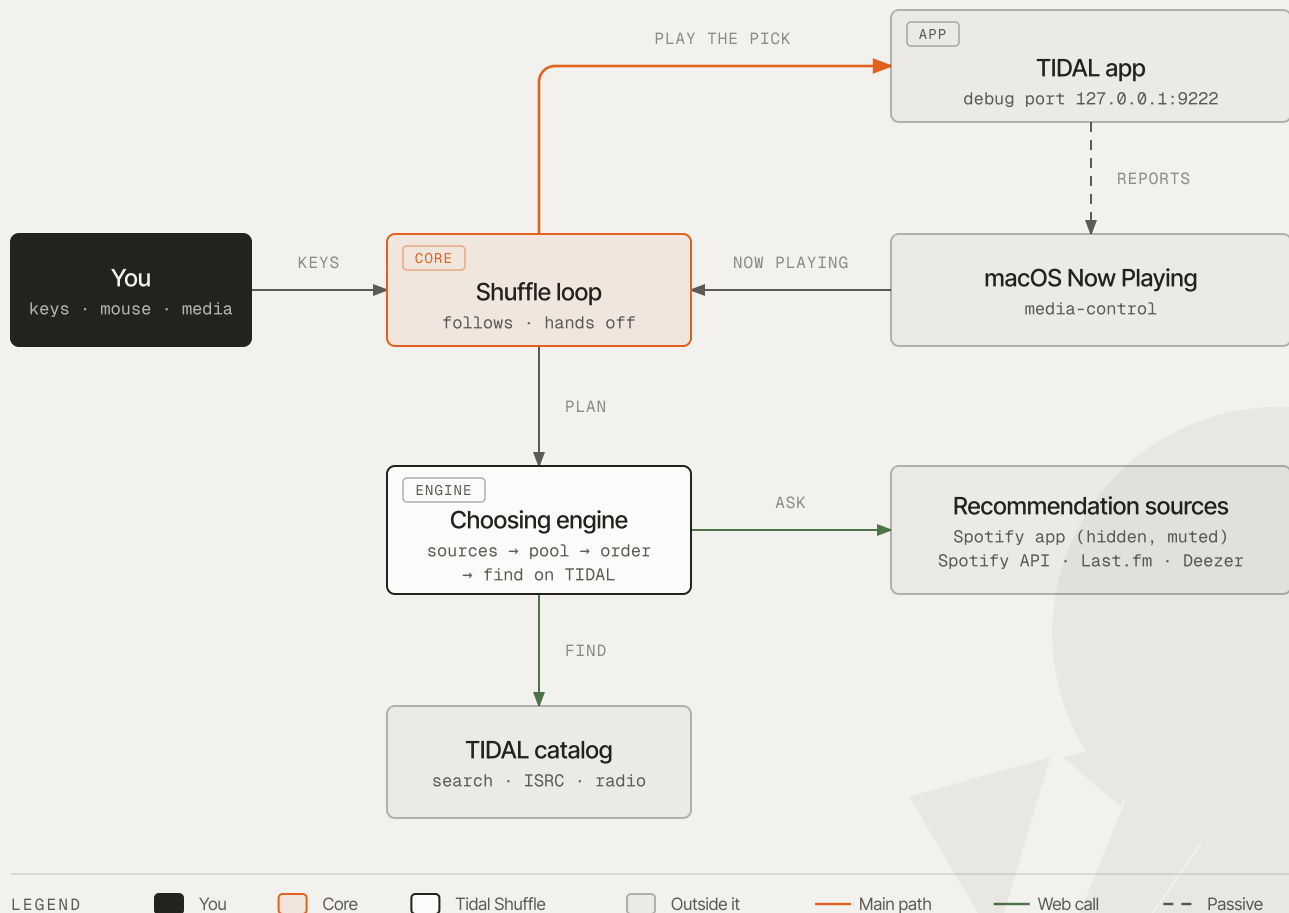


FIGURE 1 · SYSTEM OVERVIEW

The **shuffle loop** follows the song macOS reports as playing. Four seconds into each song it asks the **choosing engine** for the next one; the engine asks the recommendation sources, cleans and orders what they suggest, and finds the songs in the TIDAL catalog. Near the end of the song the loop makes the TIDAL app play the pick, through the app's local debug port.

02 · INSTALLING AND UPDATING

Installing it

NEEDS	FOR	NOTES
A Mac with the TIDAL app	everything	a TIDAL subscription; the app from tidal.com/download
Homebrew	installing	brew.sh ; the installer adds Python 3.12 and media-control with it
Spotify app	the best suggestions	free or Premium; used hidden and muted
Spotify developer app	optional	client ID and secret; its owner needs Premium (2026)
Last.fm API key	optional	another source of similar songs

2.1 Install

```
# in Terminal
git clone -b claude/blissful-pasteur-dw57cr https://github.com/canyoubarrett/vim.git ~/tidal-shuffle-src
cd ~/tidal-shuffle-src/tidal-shuffle
./install.sh
tidal-shuffle login      # approve the link it prints, in your browser
tidal-shuffle doctor    # checks every part and says how to fix what is missing
```

`install.sh` makes a private Python environment in `.venv`, installs `media-control` and puts the `tidal-shuffle` command on your path. The TIDAL login is kept in `~/.config/tidal-shuffle`.

2.2 First run

- 1 Start TIDAL and play any song.
- 2 In Terminal: `tidal-shuffle run`. If TIDAL was started without its debug port, Tidal Shuffle restarts it once, with the port (the music resumes).
- 3 The first time Spotify is used, macOS asks to let Terminal control Spotify: allow it.

2.3 Update

```
tidal-shuffle update      # fetches the latest version and reinstalls it
tidal-shuffle --version   # shows the version and where it is installed
```

Quit Tidal Shuffle first (`q`). Settings, history and pictures live in `~/.config/tidal-shuffle` and are kept. If update reports changed files, run `git -C ~/tidal-shuffle-src stash` and update again. To remove it, delete `~/tidal-shuffle-src` and `~/.config/tidal-shuffle`.

03 · OPERATING

Running it

Start it with `tidal-shuffle run` and leave the window open; it keeps the music going until you press `q`. Choose a starting style with `--preset NAME` (see 3.3), or `--plain` for a scrolling log instead of the screen below.

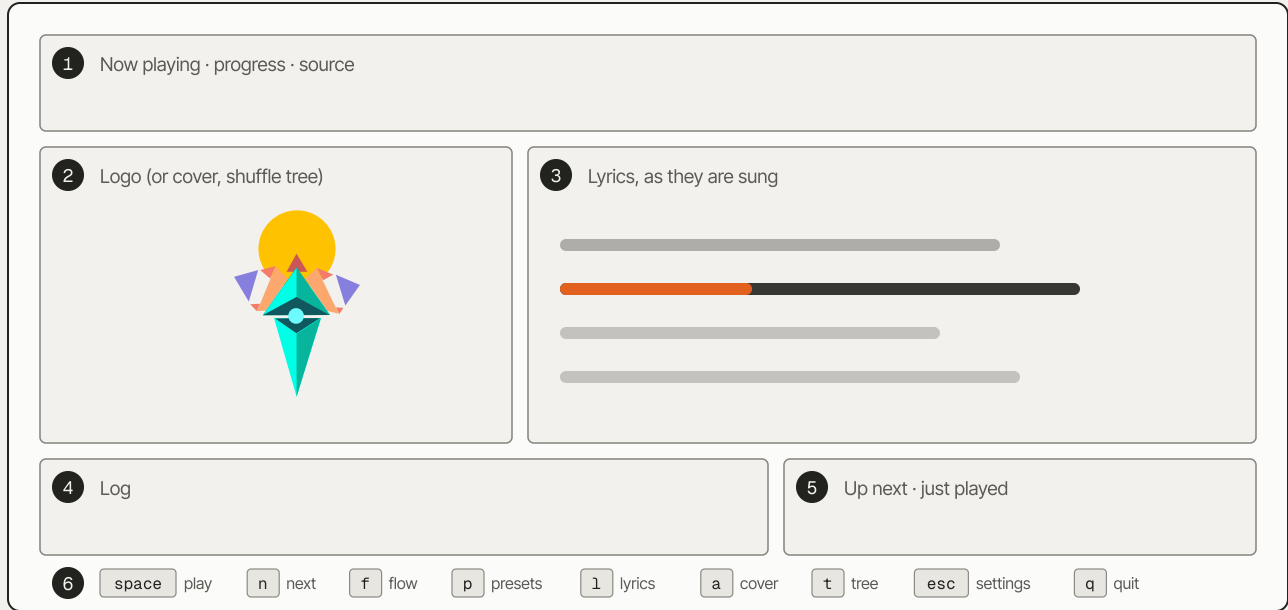


FIGURE 2 · THE FULL-SCREEN VIEW

PART	WHAT IT SHOWS
1	Header
2	Logo
3	Lyrics
4	Log
5	Up next
6	Keys

The screen adapts to the window. In a narrow window the lyrics take the whole width; Up next always stays beside the log.

03 · OPERATING

Keys and styles

3.1 Keys

KEY	DOES
<code>space</code>	play or pause TIDAL
<code>n</code>	next: start the pick now (the song after it is already being chosen)
<code>b</code>	back: the previous song in TIDAL
<code>f</code>	the next flow (how each song follows the last; see 3.3)
<code>p</code>	presets: choose a ready-made style
<code>l</code>	lyrics on or off
<code>a</code>	the album cover in place of the logo
<code>t</code>	the shuffle tree: watch the next song being chosen
<code>esc</code>	the settings menu (chapter 04)
<code>q</code>	quit; TIDAL keeps playing
<code>?</code>	help

3.2 Media keys and mouse

While the Terminal window is in front, the keyboard's `⏮` `⏭` keys work like `space` `n` `b`. macOS asks once to let Terminal see them (Accessibility). Every key in the bottom row, every preset and every setting can be clicked.

3.3 Flows

FLOW	EACH NEXT SONG IS...
radio	the song radio's own order
rising	a little more energetic
falling	a little calmer
steady	at one energy level
soundscape	close to the first song's overall sound
vibe	of the current song's mood and energy

3.4 Presets

anchor, balanced, chill, discovery, familiar, lastfm-only, late-night-drive, radio, soundscape, spotify-only, steady, tidal-only, vibe, wander, warm-up, wind-down, workout. List them with `tidal-shuffle presets`; add your own in the config file.

04 · THE SETTINGS MENU

Settings

`esc` opens the menu on the left half of the screen, with a live preview on the right. Whatever the cursor is on is shown at once; nothing is kept until you choose it.

- 1 `↑` `↓` move; `←` `→` jump a section (on a slider, they move the slider).
- 2 `enter` or a click chooses. The choice is kept between runs, in `~/ .config/tidal-shuffle/ui.json`.
- 3 `esc` closes the menu and puts back anything only previewed.

SECTION	WHAT IT SETS	KEPT AS
Theme	twenty colour themes for the whole screen (Nord, Dracula, Gruvbox, Catppuccin and more)	<code>ui.theme</code>
Party mode	the logo glides through every colour and motion	<code>ui.party</code>
Logo colours	theme, muted, the logo's own (filled or lines), pastel, neon, sunset, ocean, Catppuccin, mono	<code>ui.logo_style</code>
Logo version	lines, or flat (solid shapes, no outlines)	<code>ui.logo_version</code>
Logo detail	how finely filled colours are drawn: Low to Highest	<code>ui.logo_detail</code>
Logo motion	float, gentle, lively, shapes, tide, topple, jelly, magnet, still	<code>ui.logo_motion</code>
Logo size	slider: the logo on its own	<code>ui.logo_size</code>
Shuffle tree	off, in place of the logo, or beside it	<code>ui.shuffle_view</code>
Rain	the rain behind the panels: off, faint, soft or clear	<code>ui.rain</code>
Lyrics not sung yet	shown (default), dimmed or hidden	<code>ui.lyrics_ahead</code>
Lyrics timing	move the highlight earlier or later	<code>ui.lyrics_lead</code>
Backdrop look	sliders: zoom, dim, logo size and floor on a picture	<code>ui.backdrop_*</code>
Picture detail	Low, Medium, High, Highest, Dots (chapter 08)	<code>ui.picture_detail</code>
Logo backdrop	off, album cover, a picture each song, or one of Your pictures	<code>ui.logo_backdrop</code>
Spotify API	client ID and secret, typed or pasted; Check asks Spotify	<code>spotify.json</code>

Your pictures are folded behind one entry, Your pictures: `enter` unfolds them. A picture is loaded only when the cursor rests on it, so moving through the list stays quick.

05 · HOW THE NEXT SONG IS CHOSEN

Choosing

Every choice goes the same way: ask, clean, order, find. The shuffle tree (🔍) shows each step as it happens, with what was left out and why.

Ask the sources

Spotify radio first

30

Clean the pool

repeats, look-alikes out

24

Order, find on TIDAL

four lookups at once

6

Pick + two backups

the first three that fit

3

example run · widths are to scale

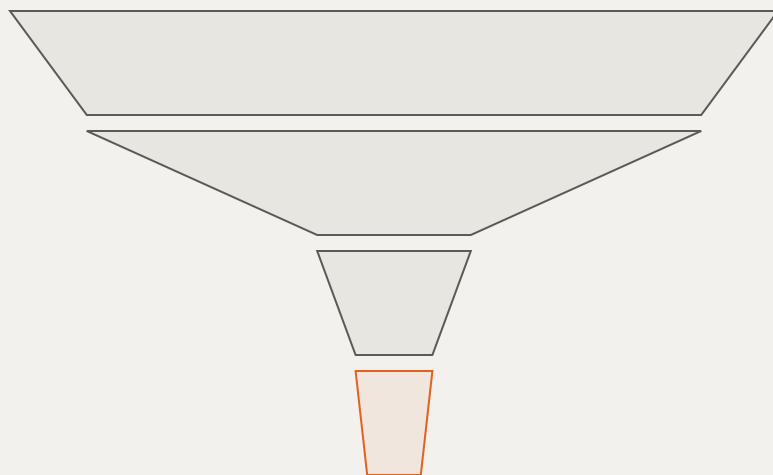


FIGURE 3 · FROM SUGGESTIONS TO A PICK (AN EXAMPLE RUN)

5.1 Ask

The sources are asked in order (`sources` in the config) until one gives enough songs. First the Spotify app: it plays the song's radio, hidden and muted, and reads the 25 songs it lines up. Then the Spotify API, Last.fm, Deezer and TIDAL's own radio.

5.2 Clean

Out go repeats, the song playing and the last few heard, songs picked in the last 200 picks or 7 days, and karaoke, tribute and lullaby versions.

5.3 Order and find

The strategy orders the pool; the flow, if any, moves songs that fit it forward. Songs are then looked up on TIDAL four at a time, but taken strictly in order, until a pick and two backups are found.

STRATEGY	PICKS
top	the best-ranked song that fits
weighted	at random, favouring the best-ranked (default)
random	any song that fits
discovery	deeper cuts and less popular songs

06 · THE HAND-OFF

Handing over to the pick

The hand-off is the moment the pick takes over from the song playing. It is timed from the song's own clock and what macOS reports; both must agree before anything happens.

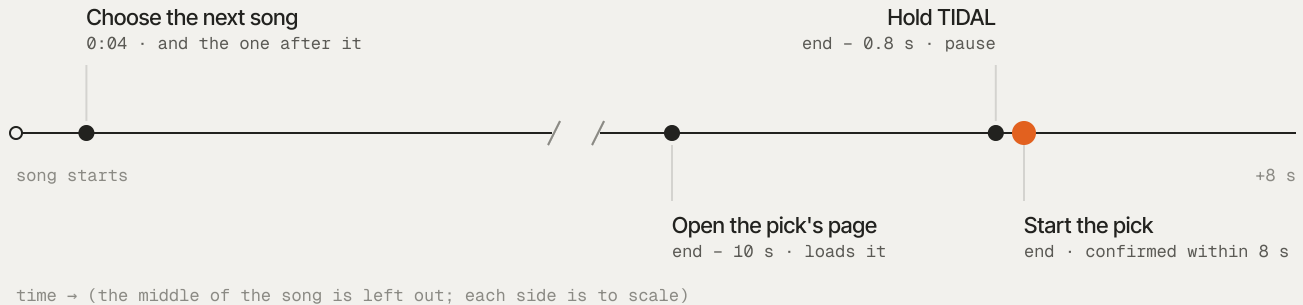


FIGURE 4 · ONE SONG, FROM ITS START TO THE HAND-OFF

6.1 Modes

MODE	AT THE END OF A SONG	USE WHEN
pause (default)	holds TIDAL 0.8 s before the end, then starts the pick	always; nothing is cut and TIDAL's own next song never plays
timed	starts the pick early, by the time TIDAL takes to start a song (learned)	you want no gap and accept a clipped end now and then
TidaLuna queue	the pick waits in TIDAL's own queue; TIDAL moves on by itself	the TidaLuna mod is installed: gapless

6.2 Choosing ahead

As soon as the pick is known, the song after it is chosen too, in the background. When the pick starts, its own next song is ready, so pressing `n` again and again never waits. If another song starts instead (you chose one in TIDAL), the early choice is dropped. Turn it off with `shuffle.plan_ahead: false`.

6.3 Starting a song in TIDAL

Tidal Shuffle opens the pick's page in the TIDAL app and presses its play button, then waits up to 8 seconds for TIDAL to show it playing. If the button cannot be found, it asks TIDAL's play queue directly, but only for a track TIDAL has loaded; opening the page loads it. After two failed picks it stops trying and says so in the log.

06 · THE HAND-OFF

Pressing next

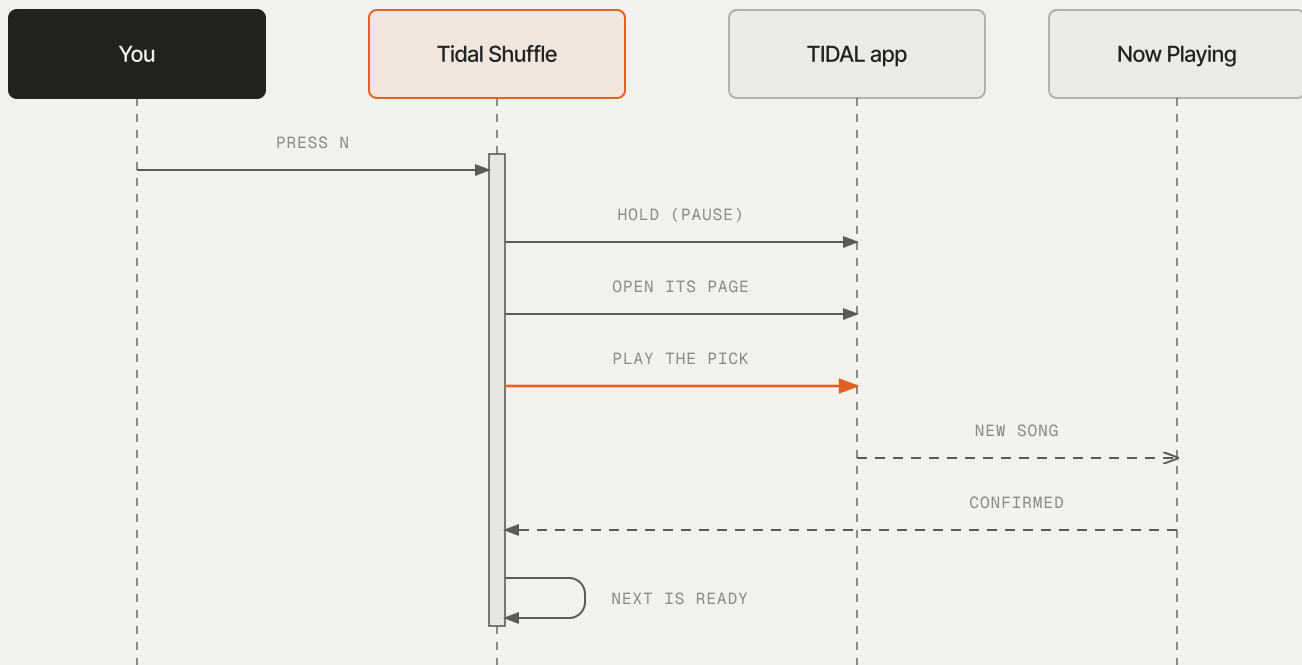


FIGURE 5 · WHAT HAPPENS WHEN YOU PRESS N

- 1 **n** reaches Tidal Shuffle (or **⏮** while its window is in front).
- 2 TIDAL is held (pause mode, the default), so you never hear TIDAL's own next song while the pick loads.
- 3 The pick's page is opened, which also loads the track in TIDAL.
- 4 The pick is played, and macOS reports the new song.
- 5 Tidal Shuffle confirms it is the pick; the song after it is already chosen.

Presses made while a switch is under way are counted once: a burst of **n** moves one song on, not five. If nothing is chosen yet, the log says "choosing a song..." and the switch follows as soon as one is.

6.4 If TIDAL moves on by itself first

Then the media key reached TIDAL too: allow Terminal under System Settings → Privacy & Security → Accessibility, and press **n** in the Tidal Shuffle window rather than **⏮** elsewhere.

07 · SOURCES AND SERVICES

Where suggestions come from

SOURCE	GIVES	NEEDS	NOTES
Spotify app	the song radio: the closest matches	the Spotify app; Automation access	hidden and muted; never interrupts your own Spotify listening
Spotify API	related artists, same-genre songs; finds Spotify ids	client ID and secret	development-mode apps need a Premium owner
Last.fm	similar tracks and artists	an API key	free
Deezer	related artists and artist radio	nothing	public API
TIDAL radio	TIDAL's track and artist radio	your TIDAL login	the fallback that is always there
ListenBrainz, Odesli	Spotify ids for TIDAL songs	nothing	used to start the Spotify radio

7.1 Adding the Spotify API

- 1 At developer.spotify.com/dashboard, create an app (any name; redirect URI not needed) and open its Settings.
- 2 In Tidal Shuffle: `esc` → **Spotify API** → **Client ID**; paste it (`⌘` `v`) and press `enter`. Then the same for the **Client secret**.
- 3 Spotify is asked at once; the answer shows beside **Check**.

The values are kept in `~/.config/tidal-shuffle/spotify.json`, readable only by you; the secret is never shown. While a field is being typed into, every key is text:

`enter` saves, `esc` cancels, `ctrl` `U` clears. The environment variables `SPOTIFY_CLIENT_ID` and `SPOTIFY_CLIENT_SECRET`, when set, take precedence.

7.2 Controlling TIDAL

The TIDAL app is controlled through its local debug port (`127.0.0.1:9222`), opened only on this Mac. Tidal Shuffle starts TIDAL with it when needed (`player.auto_relaunch`). With the optional TidaLuna mod, picks go into TIDAL's own queue for gapless changes.

08 · THE DISPLAY

The display

8.1 Colour

Terminal.app shows 256 colours: every theme colour is matched to its nearest one by eye. iTerm2, Ghostty, WezTerm and kitty show the exact colours. `tidal-shuffle colors` shows what a terminal can do; `ui.color` overrides it.

8.2 The logo

The Alter Era mark, as a line drawing or flat shapes, in the theme's colours or its own. It floats, tilts and comes apart according to its motion, faster when the music is livelier.

8.3 Backdrops

A picture behind the logo, dimmed so the logo stands out; wide ones pan slowly. On a picture the logo stands on the floor, fighter-sized. Add pictures with `tidal-shuffle backdrops add PATH` (files, folders or zips).

8.4 Picture detail

Backdrops and the large cover are drawn from the full-size picture, in steps:

STEP	PIXELS A CHARACTER	DRAWN WITH	WHERE
Low	1 × 2	half blocks	every terminal
Medium	2 × 2	quarter blocks	every terminal (default)
High	2 × 3	sixth blocks	Ghostty, kitty, WezTerm, iTerm2
Highest	2 × 4	eighth blocks	Ghostty, kitty
Dots	2 × 4	braille dots	every terminal; grainier

High and Highest use newer block characters most fonts lack; they are only offered where the terminal draws them itself (`ui.block_glyphs: all` offers them anyway). For more detail in any terminal, make its font smaller ( ).

09 · CONFIGURATION REFERENCE

The config file

Optional. `tidal-shuffle config init` writes `~/.config/tidal-shuffle/config.yaml` with every setting and a note on each; `tidal-shuffle config show` prints what is in effect. Choices made in the settings menu override the file's `ui:` section. The most used settings:

SETTING	DEFAULT	VALUES
<code>sources</code>	<code>spotify-app, spotify-api, lastfm, deezer, tidal-radio</code>	the order sources are asked in
<code>shuffle.strategy</code>	<code>weighted</code>	<code>top · weighted · random · discovery</code>
<code>shuffle.flow</code>	<code>radio</code>	<code>radio · rising · falling · steady · soundscape · vibe</code>
<code>shuffle.seed</code>	<code>current</code>	<code>current · anchor (the first song) · window (the last few)</code>
<code>shuffle.artist_cooldown</code>	<code>5</code>	picks before an artist may return
<code>shuffle.avoid_repeats_for</code>	<code>200</code>	picks before a song may return (and <code>_days</code> : 7)
<code>shuffle.lookahead</code>	<code>3</code>	the pick and its backups
<code>shuffle.plan_ahead</code>	<code>true</code>	choose the song after the pick in advance
<code>shuffle.lookup_workers</code>	<code>4</code>	TIDAL lookups at once
<code>player.handoff_mode</code>	<code>pause</code>	<code>pause · timed (6.1)</code>
<code>player.pause_before_end</code>	<code>0.8</code>	seconds before the end to hold TIDAL
<code>player.prepare_seconds</code>	<code>10</code>	seconds before the end to open the pick's page
<code>player.plan_after_seconds</code>	<code>4</code>	seconds into a song before choosing
<code>player.verify_seconds</code>	<code>8</code>	how long TIDAL has to confirm the pick
<code>player.media_keys</code>	<code>focus</code>	<code>focus (window in front) · always · off</code>
<code>spotify.app.harvest</code>	<code>25</code>	radio songs read per seed
<code>ui.theme</code>	<code>mocha</code>	any of the twenty themes
<code>ui.color</code>	<code>auto</code>	<code>auto · truecolor · 256 · 16</code>
<code>ui.picture_detail</code>	<code>quadrant</code>	<code>half · quadrant · sextant · octant · braille</code>
<code>ui.block_glyphs</code>	<code>auto</code>	<code>auto · all · basic</code>
<code>ui.fps</code>	<code>12</code>	frames a second; lower it on a slow Mac

10 · TROUBLESHOOTING

Putting it right

Start with `tidal-shuffle doctor`: it checks every part (macOS, TIDAL, the login, the debug port, media-control, Spotify, the sources) and says how to fix each one it marks.

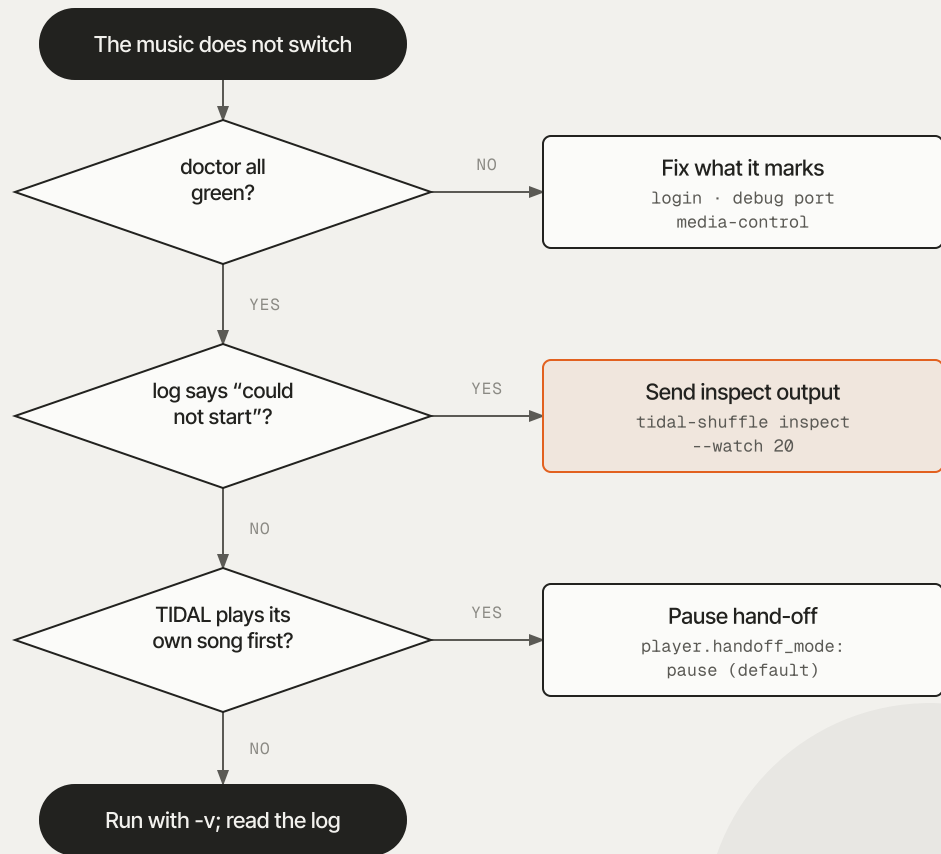


FIGURE 6 · WHEN THE MUSIC DOES NOT SWITCH

SYMPTOM	CAUSE	REMEDY
TIDAL plays its own next song	the media key reached TIDAL too	allow Terminal in Accessibility; press n in the window
"could not start" in the log	TIDAL's page has changed	<code>tidal-shuffle inspect --watch 20</code> ; send the output
Nothing is chosen	no source answered	<code>tidal-shuffle sources</code> shows each source's answer
No colours	the terminal, or <code>NO_COLOR</code> set	<code>tidal-shuffle colors</code> ; <code>ui.color: 256</code>
Boxes with question marks	the font lacks the characters	Esc → Picture detail → Medium
Spotify keeps appearing	its search needed the window	<code>spotify.app.keep_hidden: true</code>

11 · FILES, COMMANDS AND REVISIONS

Reference

11.1 Files, in ~/.config/tidal-shuffle

FILE	HOLDS
config.yaml	the config file (optional)
ui.json	choices made in the settings menu
spotify.json	Spotify API credentials (readable only by you)
tidal_session.json	the TIDAL login
history.json	songs played and picked
backdrops/	your pictures
cache/	lyrics, covers, ids; safe to delete
timing.json	how long TIDAL takes to start a song

11.2 Commands: tidal-shuffle ...

COMMAND	DOES
run	follow TIDAL and keep the music going
next	skip to a fresh pick now
doctor	check every part
update	get the latest version
login	log in to TIDAL
test	show what would play next, without playing
sources	ask every source once
presets	list the presets
history	show or clear the history
backdrops	add or list pictures
colors	show what the terminal can display
inspect	dump TIDAL's page, for fixing
config	create or show the config file

11.3 Revisions

REV	DATE	CHANGES
0.12.0	3 Oct 2026	the song after the pick chosen ahead; lookups four at a time; Spotify API in settings; pictures folded away
0.11.1	3 Oct 2026	Up next beside the log; upcoming lyrics shown; no boxes in Terminal.app
0.11.0	3 Oct 2026	next starts the pick without TIDAL's own next; finer detail; the flat logo

Alter Era · Engineering. Tidal Shuffle is not made by or affiliated with TIDAL or Spotify.